

3.3 → Special case: Limited

3.3.2 → Incremental limits reinsurance

- Incremental limits reinsurance is the process of determining individual allocated limit factors (ILFs)
- why needed? → limited data leads to reliable results. multivariate approaches are better for sparse data, but can produce illogical rating factors (e.g. decrease in limit for a higher limit)
- have special case approaches

→ Standard ILF approach

$$\text{Rate}_H = \text{ILF} \times \text{Rate}_0$$

$$\text{Individual ILF} = \frac{\text{LIS}(H)}{\text{LIS}(0)}$$

- Assumptions:
 - all underwriting exposures are variable
 - the variable exposure & limit conditions do not vary by limit
 - frequency & severity are independent
 - frequency is the same for all limits

→ LIS(H), or the limited exposed severity < H, is the severity assuming every unit is exposed at limit H regardless of an actual policy limit. LIS(0), or the limited severity at 0, is the severity assuming every unit is capped at the base limit.

$$\text{LIS}(H) = \frac{\text{Total loss at limits} < H}{\text{Total claims}} + \frac{H \times \text{Claims} > H}{\text{Total claims}}$$

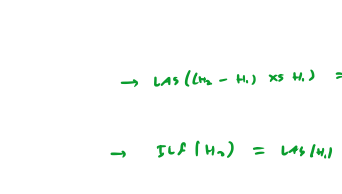
→ Censored losses → for $H_0 > H_1$

$$\text{LIS}(H_0) = \text{LIS}(H_1) \times \frac{H_0 \times \text{Claims} > H_1}{H_1 \times \text{Claims} > H_1}$$

$$\text{LIS}(H_0) = \text{LIS}(H_1) \times \frac{H_0}{H_1}$$

→ Noticing that the numerator of LIS(H₁) equals the denominator of LIS(H₀) × H₁ allows us to form a fraction:

$$\text{LIS}(H_0) = \text{LIS}(H_1) + \frac{H_0 \times \text{Claims} > H_1}{H_1 \times \text{Claims} > H_1}$$



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3.3.3 → Deductible policies

- Deductible policies are a common type of deductible: fixed dollar or percentage
- Deductibles are popular for:
 - premium reduction (from insured's perspective)
 - Eliminate small deductible claims
 - Provide incentive for loss control (financial incentive for insured to prevent losses since they are partially responsible for paying)
 - Control catastrophe exposure

→ Deductible reinsurance, follows from the loss elimination ratio (LER) approach

→ Assuming all exposures are variable & the variable exposure & limit are a constant percentage of premium, the insured deductible reinsurance for given limit and retention level can be calculated as:

$$\text{Insured deductible reinsurance} = \frac{(L+R)_0}{(L+R)_D} = \frac{L+R}{L+R} = 1 - \text{LER}(D)$$

where $(L+R)_0$ & $(L+R)_D$ are the limit & LER net of the deductible D & D , respectively.

$$\text{LER}(D) = \frac{(L+R)_0 - (L+R)_D}{(L+R)_0}$$

→ Approximating the proportion of losses eliminated by going from one deductible to another. Greatly simplified, the LER elimination ratio is an estimate of the proportion of losses that remain after deducting a deductible.

→ Paths → for policies of full coverage, for $D=0$, then the formula simplifies to:

$$\text{Insured deductible reinsurance} = \frac{\text{Limit} + \text{LER} \text{ assuming deductible } D}{\text{Total gross limit} + \text{LER}}$$

$$\text{LER}(D) = \frac{\text{Limit} + \text{LER} \text{ assuming deductible } D}{\text{Total gross limit} + \text{LER}}$$

→ Two methods to calculate LER:

→ Method 1 → The ratio of losses eliminated by the deductible to the total gross-up losses

→ Method 2 → One minus the ratio of the losses that remain after applying the deductible to the gross-up losses

→ Not reported losses → the portion of reported losses that exceed the deductible

→ When values are losses (not expenses), both from policies and higher deductibles cannot be used to calculate the LER for lower deductibles.

→ Both from same deductible cannot be used to calculate the LER for a 0 deductible.

But the ratios work!

Limit: 500000 → 500000

Limit: 500000 → 500000

→ Fitted data approach

→ Assume also denotes the continuous distribution of losses at size x & D is the size of the deductible. Assuming the deductible is the base limit policy, the limit eliminated can be calculated as:

$$\text{Eliminated loss} = \int_0^D x f(x) dx + \int_D^{\infty} D f(x) dx \Rightarrow \text{if } f(x) = \frac{x}{D} \times e^{-x/D} \Rightarrow E(x) = D$$

→ Then the LER is the eliminated loss divided by the expected loss:

$$\text{LER}(D) = \frac{\text{Eliminated loss}}{\text{Expected loss}}$$

$$\text{LER}(D) = \frac{\int_0^D x f(x) dx + \int_D^{\infty} D f(x) dx}{\int_0^{\infty} x f(x) dx}$$

→ Practical considerations → the following should be considered when using the LER approach:

→ LER calculation assumes eliminating exposure will be the same for each deductible & across the deductibles

→ e.g. under 50000, a claim of 50000 will likely be reported, but for a 500000, that same loss may not be reported as the payment of 50000 doesn't outweigh the premium payments for having a claim

→ LER does not restrict the behavioral differences that arise when insureds are allowed to choose their policy deductible

→ e.g. low risk insureds → often choose high deductibles → often higher risk insureds are more profitable for insurers than lower risk insureds (better ability to risk transfer)

→ insurance & reinsurance AGS able to capture these behavioral differences

→ Good idea to use multiple methods when calculating LER

→ Premium credits → If LER for a 50000 deductible is say 75% → 75% of the total losses will be eliminated by moving to the deductible

→ say the policy premium is 50000, then that should equal to a premium credit of 50000 × 0.75 = 37500, but

→ But, for a much larger than the deductible (which is the amount of moving the insurer pays for policy (premium credit))

→ Insurer can remedy this by:

→ capping the credit at say 37500

→ Calculating different set of credits for different policies (e.g. different credits for low, medium, or high risk insureds), Segregating risks based on the likelihood of such remedies

→ Use the deductibles, which allow the credit to vary by coverage amount

3.3.4 → Losses Compensation Size at Risk

→ ... maybe read? ...

3.3.5 → Insurance to Value

→ Insurance to value (ITV) describes the relationship between the claim amount of reinsurance & the total value or replacement cost of the insured.

→ But it what insurers should share risk on to ensure equitable & adequate rates

→ Issue that arise when properties (for homeowning insurance) are not insured to their full value of replacement cost

→ 1) The replacement payment will not adequate to cover a total or near-total loss, leaving the insured unable to fully recover their property

→ 2) The limit is higher than 50000, but only insured for 50000, so loss to 50000 will be covered, which leaves the insured with a gap

→ 3) So the insured has to risk on the assumption that all losses are insured to their full value, the premium charged will not be sufficient to cover the expected loss or underinsured status

→ Importance of ITV

→ When properties are not insured to the same level → rate inequity

→ But it is all homes are underinsured by the same percentage → the resulting premium may not be adequate to cover all losses, but it would be equitable

→ over time, the insurer will raise the rates so that the appropriate premium covers the expected aggregate losses.

→ However, in these scenarios, fully insured homes end up bearing part of the cost of the underinsured homes.

→ ITV is important to note that the rates of inequity & inadequacy of rates only exist if partial losses are possible. If all losses are total losses, the total premium collected will be adequate to cover all losses, & rates will be equitable, even if some homes are underinsured

→ Therefore, lower losses are partial rather than total. To ensure rates that are equitable & adequate, insurers should charge rates based on the ITV level.

→ other ITV calculations

→ other government-backed reinsurance cost (e.g. reinsurance of a cap)

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